

CHAPTER 5 - CLUSTERING WITH BAYES

LECTURER: RICCARDO CORRADIN

UNIVERSITY OF MILANO-BICOCCA

1 INTRODUCTION TO CLUSTERING

Clustering is one of the fundamental techniques in statistical analysis, ideally to handle situations where data shows heterogeneity. Suppose we have a sequence $y_1, \dots, y_n$ of observations, where the generic $y_i \in \mathbb{Y}$. In cluster analysis, we aim to identify subsets of observations which result to be similar in their observed values. With refer to similarity of observations the case where observations within a specific group are somehow homogeneous. The specific definition depends on which model we are assuming and what measures similarity among observations.

We remark that, differently from the mixed model of the previous chapter, here the groups are unknown and object of inference, hence we are falling in the unsupervised learning framework. We do not know a priori how observations are grouped with each other, and we want to infer a partition structure that divide our observations in homogeneous groups.

In the following sections, we assume that $\mathbb{Y} \subseteq \mathbb{R}^p$, i.e. we aim to produce homogeneous clusters of real-valued quantities. However, many generalizations can be considered. In simple scenarios, defining clusters can be quite trivial. However, when the number of dimensions increases, or the domain is getting complex, we should be more careful in the specification of our model.

Clustering techniques can be roughly divided in two macro-categories, dictated by their way of describing similarities among observations. In model-based clustering, where we assume the observations to be distributed according to cluster-specific distributions. Hence, the homogeneity among observations is driven by a probabilistic models.

- Quite flexible approach, the group-specific distribution can be any kind of model and it plays the role of the group likelihood.
- The group-specific distribution can be useful also to interpret the data behavior.
- In this case, we call the joint model “mixture model” (different from the mixed model of the previous chapter).

Alternative, with Distance-based clustering we use a distance to measure dissimilarities among observations, from a centroid of interest, and the clusters are defined by observation closed to each others.

- There are several distances that we can use, depending on specific problems.
- There is no interpretation in terms of model structure.
- We do not have an usual likelihood function.

Here, we will study how to perform cluster analysis with a Bayesian approach in the first case, where we assume a cluster-specific distribution. The key objects to perform model-based clustering are the so-called mixture models. A mixture model over $\mathbb{Y}$ is nothing but a weighted average of different density functions or probability mass functions, having component-specific parameter values, of the form

$$f(\mathbf{y}) = \sum_{j=1}^k w_j k(\mathbf{y}, \boldsymbol{\theta}_j^*),$$

where $\boldsymbol{\theta}_j^* \in \Theta \subseteq \mathbb{R}^q$, $j = 1, \dots, k$, and $k \in \mathbb{N} \cup \{\infty\}$, and the kernel function $k(\cdot, \cdot)$ is shared across components. Specifically, the sequence $w_1, \dots, w_k$ is a sequence of non-negative weights taking values in the $(k - 1)$ -dimensional simplex space Δ_{k-1} , i.e.

$$\Delta_{k-1} = \left\{ w_1, \dots, w_k : 0 \leq w_j \leq 1, j = 1 \dots, k, \sum_{j=1}^k w_j = 1 \right\}.$$

$k(y, \theta)$ is a kernel function satisfying the following regularity conditions: (i) $k(\cdot, \theta)$ is a density function or a probability mass function for any value of $\theta \in \Theta$; (ii) $k(y, \cdot)$ is measurable for any value of $\mathbf{y} \in \mathbb{Y}$.

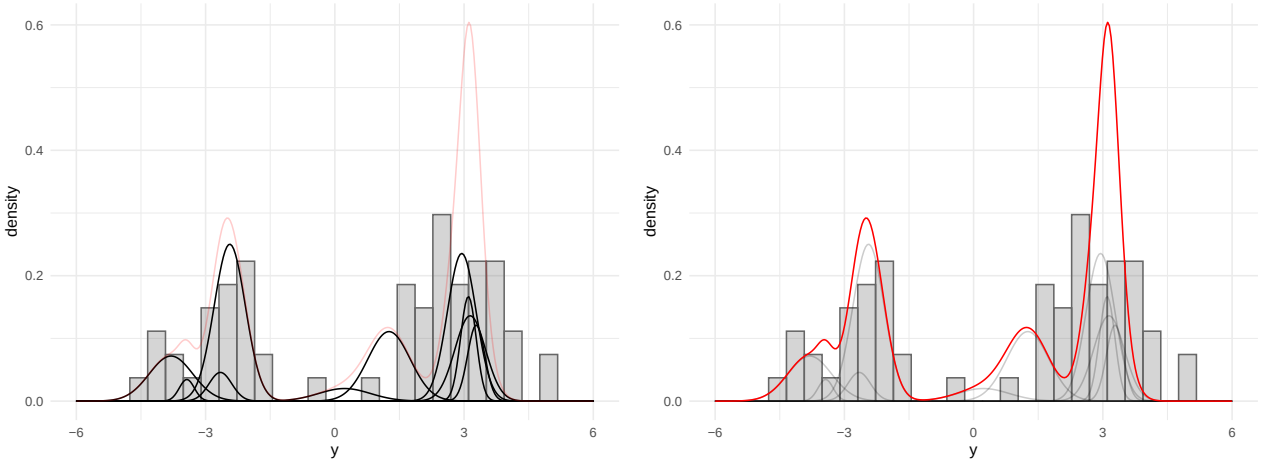


Figure 1: Left panel: different components of the mixture model. Right panel: a single mixture model.

Mixture models are flexible and can capture many complex distributional behavior, such as multimodality and skewness, while preserving a simple form of group-specific components. They are naturally tailored to perform model based clustering, as each component can represent a single cluster, defining equivalence classes among observations. Despite their simple specification, they can accommodate many different type of data by suitably specifying the kernel function.

- $k(\mathbf{y}, \boldsymbol{\theta}) \stackrel{d}{=} N(\mu, \sigma)$ to cluster univariate data defined on $\mathbb{R}$.
- $k(\mathbf{y}, \boldsymbol{\theta}) \stackrel{d}{=} N(\boldsymbol{\mu}, \Sigma)$ to cluster univariate data defined on $\mathbb{R}^d$.
- $k(\mathbf{y}, \boldsymbol{\theta}) \stackrel{d}{=} GP(\mu(t), R(t, t'))$ to cluster functional data taking values on $C_{\mathbb{X}}$, space of continuous function with support $\mathbb{X}$.

- $k(\mathbf{y}, \boldsymbol{\theta}) \stackrel{d}{=} \text{ERGM}(\boldsymbol{\theta})$ to cluster graph data.

- ...

All the benefits come at a cost. The estimation is usually computationally intensive, especially with complex data, and we require suitable computational strategies to produce posterior inference. In particular, they might be prohibitive with large sample sizes, as usual inferential procedure are not scaling well with the sample size. Further, here we consider the case of finite mixture models, with a finite number of components. Finite mixture models can be sensible to their specification, especially regarding the number of components.

To complete the model specification in a Bayesian fashion, we need to assume a specific weights distribution, which describes a priori how we allocate mass on the components, i.e. $\mathbf{w} \sim \pi(\mathbf{w})$, a kernel function $k(\mathbf{y}, \boldsymbol{\theta})$, which depends on the specific data we are analyzing, and a distribution for the cluster-specific parameters $\boldsymbol{\theta}_j^*$, that usually depends on the kernel function we assume and it is diffuse on the parameter support.

One common choice for the weights distribution is the Dirichlet distribution, a distribution taking support on simplex spaces. Specifically, we say $\mathbf{w} \sim \text{Dirichlet}(\boldsymbol{\alpha})$ if the distribution of $\mathbf{w}$ is characterized by the following density function

$$\pi(\mathbf{w}) = \frac{\Gamma(\alpha_1 + \dots + \alpha_k)}{\Gamma(\alpha_1) \dots \Gamma(\alpha_k)} \prod_{j=1}^k w_j^{\alpha_j-1}, \quad \mathbf{w} \in \Delta_{k-1}^1.$$

Recall that $\mathbb{E}[W_j] = \frac{\alpha_j}{\alpha^+}$ and $\text{Var}(W_j) = \frac{\alpha_j(\alpha^+ - \alpha_j)}{\alpha^+(\alpha^+ + 1)}$, where $\alpha^+ = \sum_{j=1}^k \alpha_j$. Different choices of $\alpha_1, \dots, \alpha_k$ lead to different specification. The case $\alpha_1 = \dots = \alpha_k = \alpha$ is called symmetric, and corresponds to have a fair prior guess across mixture components, since they have the same prior probability. In the symmetric case, common settings are $\alpha = 1$ or $\alpha = \frac{1}{k}$. The first corresponds to an uniform prior over Δ_{k-1}^1 , while the second introduce a penalization on the number of the mixture components. Finally, the distribution of $\boldsymbol{\theta}_j^*$, denoted by $\pi(\boldsymbol{\theta}_j^*)$, depends on the kernel assumption. Usually, to improve the model tractability, we prefer prior distribution that are conjugate to the kernel function.

Under the previous model settings, our observed data are distributed according to a mixture model, i.e. each observation is sampled from the weighted average of kernel functions defined before

$$\begin{aligned} \mathbf{y}_1, \dots, \mathbf{y}_n \mid \mathbf{w}, \boldsymbol{\theta}_{1:k}^* &\stackrel{iid}{\sim} f(y) = \sum_{j=1}^k w_j k(\mathbf{y}, \boldsymbol{\theta}_j^*) \\ \mathbf{w} &\sim \text{Dir}(\boldsymbol{\alpha}), \\ \boldsymbol{\theta}_1^*, \dots, \boldsymbol{\theta}_k^* &\stackrel{iid}{\sim} \pi(\boldsymbol{\theta}), \end{aligned}$$

where $\pi(\boldsymbol{\theta})$ is a suitable prior for the component-specific parameter, matching its support.

Remark 1.1. In the current model specification, the distribution of the weights $\mathbf{w}$ and the atoms $\boldsymbol{\theta}_{1:k}^*$ are independent. Hence, cluster-specific behaviors are independent of their probabilities. Further, the cluster-specific parameters $\boldsymbol{\theta}_{1:k}^*$ are i.i.d. from a common distribution $\pi(\boldsymbol{\theta})$. The behavior of a cluster is not affecting other clusters. Both the previous can be relaxed extending mixture models to more flexible scenarios.

Despite its flexibility, within this formulation the model is not particularly tractable, since the likelihood function becomes

$$L(\mathbf{y}_{1:n} \mid \mathbf{w}, \boldsymbol{\theta}_{1:k}^*) = \prod_{i=1}^n \left(\sum_{j=1}^k w_j k(\mathbf{y}_i, \boldsymbol{\theta}_j^*) \right).$$

In practice, our posterior distribution is proportional to the product of the previous likelihood and the prior distribution assumptions, i.e.

$$\pi(\mathbf{w}, \boldsymbol{\theta}_{1:k}^* \mid \mathbf{y}_{1:n}) \propto L(\mathbf{y}_{1:n} \mid \mathbf{w}, \boldsymbol{\theta}_{1:k}^*) \pi(\mathbf{w}) \prod_{j=1}^k \pi(\boldsymbol{\theta}_j^*),$$

and we do not recognize any choices of $\pi(\mathbf{w})$ and $\pi(\boldsymbol{\theta}_j^*)$ that lead to posterior tractability. Furthermore, our interest relies in clustering procedure, hence we want to induce an allocation of the observation to specific component of the mixture, to perform clustering analysis. We can augment the model space with suitable quantities.

1.1 A FIRST MODEL AUGMENTATION

As commonly done when the problem is hardly tractable, we consider a data augmentation that somehow leads to posterior tractability. We can choose among two augmentations, with different interpretations. At first, we can consider a sequence of latent parameters $\boldsymbol{\theta}_1, \dots, \boldsymbol{\theta}_n$, each associated to a specific observations, having

$$\begin{aligned} \mathbf{y}_i \mid \boldsymbol{\theta}_i &\sim k(\mathbf{y}_i, \boldsymbol{\theta}_i), \quad i = 1, \dots, n \\ \boldsymbol{\theta}_1, \dots, \boldsymbol{\theta}_n \mid \mathbf{w}, \boldsymbol{\theta}_{1:k}^* &\stackrel{iid}{\sim} \sum_{j=1}^k w_j \delta_{\boldsymbol{\theta}_j^*}(\boldsymbol{\theta}), \\ \mathbf{w} &\sim Dir(\boldsymbol{\alpha}), \\ \boldsymbol{\theta}_1^*, \dots, \boldsymbol{\theta}_k^* &\stackrel{iid}{\sim} \pi(\boldsymbol{\theta}). \end{aligned}$$

Proposition 1.2. *Marginalizing with respect to $\boldsymbol{\theta}_{1:n}$ the joint augmented likelihood $L(\mathbf{y}_i, \boldsymbol{\theta}_i \mid \mathbf{w}, \boldsymbol{\theta}_{1:k}^*)$ we recover the likelihood function $L(\mathbf{y}_i \mid \mathbf{w}, \boldsymbol{\theta}_{1:k}^*)$.*

Proof. The augmented likelihood term is given by

$$L(\mathbf{y}_{1:n}, \boldsymbol{\theta}_{1:n} \mid \mathbf{w}, \boldsymbol{\theta}_{1:k}^*) = \prod_{i=1}^n k(\mathbf{y}_i, \boldsymbol{\theta}_i) \sum_{j=1}^k w_j \delta_{\boldsymbol{\theta}_j^*}(\boldsymbol{\theta}_i)$$

We then have

$$\begin{aligned} \int_{\Theta^n} \prod_{i=1}^n k(\mathbf{y}_i, \boldsymbol{\theta}_i) \sum_{j=1}^k w_j \delta_{\boldsymbol{\theta}_j^*}(\boldsymbol{\theta}_i) d\boldsymbol{\theta}_1 \cdots d\boldsymbol{\theta}_n &= \int_{\Theta^n} \prod_{i=1}^n \sum_{j=1}^k w_j k(\mathbf{y}_i, \boldsymbol{\theta}_i) \delta_{\boldsymbol{\theta}_j^*}(\boldsymbol{\theta}_i) d\boldsymbol{\theta}_1 \cdots d\boldsymbol{\theta}_n \\ &= \prod_{i=1}^n \int_{\Theta} \sum_{j=1}^k w_j k(\mathbf{y}_i, \boldsymbol{\theta}_i) \delta_{\boldsymbol{\theta}_j^*}(\boldsymbol{\theta}_i) d\boldsymbol{\theta}_i = \prod_{i=1}^n \sum_{j=1}^k w_j k(\mathbf{y}_i, \boldsymbol{\theta}_j^*) \end{aligned}$$

which corresponds to the quantity of interest. □

In practice, we are matching each observation with a specific latent parameter, under the distributional assumption

$$\theta_1, \dots, \theta_n \mid \mathbf{w}, \theta_{1:k}^* \stackrel{iid}{\sim} \sum_{j=1}^k w_j \delta_{\theta_j^*}(\theta).$$

The previous is an example of random probability measure, since it is a probability measure itself, but both weights $\mathbf{w}$ and atoms $\theta_{1:k}^*$ are random quantities. Further, the previous is a discrete random probability measure. From its discreteness, the sequence of latent parameters $\theta_1, \dots, \theta_n$ can then have ties, meaning that

$$P(\theta_i = \theta_j) > 0, \quad i \neq j.$$

Ties implies that the observations can be grouped together when the corresponding latent parameters take the same unique values θ_j^* , hence they are generated from the same kernel function. This is naturally inducing a clustering within the data, where clusters mean that observations share the same distribution in a model-based case. Note that this is not happening with a generic diffuse prior distribution for the θ_i 's, since the probability that two latent parameters coincide in this case became zero.

1.2 A SECOND MODEL AUGMENTATION

While the previous augmentation has a nice interpretation, in practice is more convenient to work with another strategy. A second augmentation can be define considering a sequence of latent indicators $s_1, \dots, s_n$, one for each observation, describing which component is the observation associated to. This augmented variables are simply distributed as a categorical random variable, having probability mass function

$$p(s_i \mid \mathbf{w}) = \prod_{j=1}^k w_j^{\mathbf{1}_{[s_i=j]}}.$$

Hence, the augmented model specification corresponds to

$$\begin{aligned} \mathbf{y}_i \mid s_i, \theta_{1:k}^* &\sim k(\mathbf{y}_i, \theta_{s_i}^*), \quad i = 1, \dots, n \\ s_1, \dots, s_n \mid \mathbf{w} &\stackrel{iid}{\sim} \text{Cat}(w_1, \dots, w_k), \\ \mathbf{w} &\sim \text{Dir}(\boldsymbol{\alpha}), \\ \theta_1^*, \dots, \theta_k^* &\stackrel{iid}{\sim} \pi(\theta). \end{aligned}$$

Two observations i and j , with $i \neq j$, are in the same cluster if they are associated to the same components, i.e. if $s_i = s_j$. This different augmentation, even if cannot be interpreted in terms of latent parameter, is quite useful to perform posterior inference with these models.

Proposition 1.3. *Also for this second augmentation, we can easily see that marginalizing with respect to $s_{1:n}$ the joint distribution of $\mathbf{y}_{1:n}, s_{1:n} \mid \mathbf{w}, \theta_{1:k}^*$, we recover the likelihood function $L(\mathbf{y}_{1:n} \mid \mathbf{w}, \theta_{1:k}^*)$.*

Proof. The augmented likelihood takes form

$$L(\mathbf{y}_{1:n}, s_{1:n} \mid \mathbf{w}, \theta_{1:k}^*) = \prod_{i=1}^n k(\mathbf{y}_i, \theta_{s_i}^*) \prod_{j=1}^k w_j^{\mathbf{1}_{[s_i=j]}}.$$

Hence, for each observation we have

$$\begin{aligned} \sum_{s_i=1}^k L(\mathbf{y}_i, s_i \mid \mathbf{w}, \boldsymbol{\theta}_{1:k}^*) &= \sum_{s_i=1}^k \prod_{j=1}^k k(\mathbf{y}_i, \boldsymbol{\theta}_{s_i}^*) w_j^{\mathbf{1}_{[s_i=j]}} \\ &= \sum_{j=1}^k w_j k(\mathbf{y}_i, \boldsymbol{\theta}_j^*) \end{aligned}$$

which corresponds to the quantity of interest. $\square$

Clusters are naturally defined, since two observations belong to the same cluster if they share the same augmented allocation value, i.e. if $s_i = j$ and $s_\ell = j$, then both observations i and ℓ belong to the j th cluster.

1.3 SAMPLING FROM THE POSTERIOR

With the previous model setting, we are able to derive the full conditional distributions of the weights $\mathbf{w}$, the group-specific parameters $\boldsymbol{\theta}_{1:k}^*$, and the augmented variables. Let us assume a mixture model with a generic kernel function $k(\mathbf{y}, \boldsymbol{\theta})$, a Dirichlet prior for the weights $\mathbf{w} \sim \text{Dir}(\boldsymbol{\alpha})$, and a suitable prior for the group-specific parameters $\pi(\boldsymbol{\theta})$. Further, we resort to the second augmentation strategy for mixture models.

Proposition 1.4. *Under the second augmentation, we have*

$$\begin{aligned} s_i \mid \mathbf{y}_i, \mathbf{w}, \boldsymbol{\theta}_{1:k}^* &\stackrel{\text{ind}}{\sim} \text{Cat}(w_1 k(\mathbf{y}_i, \boldsymbol{\theta}_1^*), \dots, w_k k(\mathbf{y}_i, \boldsymbol{\theta}_k^*)), \quad i = 1, \dots, n, \\ \mathbf{w} \mid s_{1:n} &\sim \text{Dir}(\alpha_1 + n_1, \dots, \alpha_k + n_k), \\ \boldsymbol{\theta}_j^* \mid s_{1:n}, \mathbf{y}_{1:n} &\stackrel{\text{iid}}{\sim} \pi(\boldsymbol{\theta}_j^* \mid s_{1:n}, \mathbf{y}_{1:n}) \propto \pi(\boldsymbol{\theta}_j^*) \prod_{i:s_i=j} k(\mathbf{y}_i, \boldsymbol{\theta}_j^*), \quad j = 1, \dots, k, \end{aligned}$$

where $n_j = \sum_{i=1}^n \mathbf{1}_{[s_i=j]}$ is the number of observations assigned to the j th component of the mixture.

Proof. Breaking down the augmented likelihood, we have

$$\begin{aligned} L(\mathbf{y}_{1:n} \mid s_{1:n}, \boldsymbol{\theta}_{1:k}^*) &= \prod_{j=1}^k \prod_{i:s_i=j} k(\mathbf{y}_i, \boldsymbol{\theta}_j^*), \\ p(s_{1:n} \mid \mathbf{w}) &= \prod_{i=1}^n \prod_{j=1}^k w_j^{\mathbf{1}_{[s_i=j]}}. \end{aligned}$$

A priori we have

$$\begin{aligned} \pi(\mathbf{w}) &= \frac{\Gamma(\alpha_1 + \dots + \alpha_k)}{\Gamma(\alpha_1) \dots \Gamma(\alpha_k)} \prod_{j=1}^k w_j^{\alpha_j - 1}, \\ \pi(\boldsymbol{\theta}_{1:k}^*) &= \prod_{j=1}^k \pi(\boldsymbol{\theta}_j^*). \end{aligned}$$

Hence, the full conditional for the weights $\mathbf{w}$ is given by

$$\pi(\mathbf{w} \mid s_{1:n}) \propto \prod_{j=1}^k w_j^{\alpha_j-1} \prod_{i=1}^n w_j^{\mathbf{1}_{[s_i=j]}} = \prod_{j=1}^k w_j^{\alpha_j+n_j-1}$$

which identifies a Dirichlet distribution. Regarding the group-specific parameters $\theta_{1:k}^*$, we have

$$\pi(\theta_j^* \mid s_{1:n}, \mathbf{y}_{1:n}) \propto \pi(\theta_k^*) \prod_{i:s_i=j} k(\mathbf{y}_i, \theta_j^*),$$

where some simplifications apply for suitable choices of the kernel function and the prior $\pi(\theta_j^*)$. Finally, for the augmented categorical variable describing the group allocation, we have

$$P(s_i = j \mid \mathbf{y}_i, \mathbf{w}, \theta_{1:k}^*) \propto w_j^{\mathbf{1}_{[s_i=j]}} k(\mathbf{y}_i, \theta_j^*),$$

which, as j varies over $1, \dots, k$, describe a categorical distribution

$$s_i \sim \text{Cat}(w_1 k(\mathbf{y}_i, \theta_1^*), \dots, w_k k(\mathbf{y}_i, \theta_k^*)).$$

□

The previous can guide us in the specification of the prior distributions parameters, since we have a clear view of how the parameters impact the posterior behavior. In particular, we have the following guidelines.

- The Dirichlet distribution should be symmetric, in force of the label switching, whereas each α_j can be interpreted as the prior sample sizes of the generic j th cluster, i.e. how much we weight the prior information within each cluster.
- The prior distribution $\pi(\theta_j^*)$ depends on the specific kernel function, but usually is set to be vague or noninformative.

Ideally, we have a latent partition induced by ties among latent allocations or parameter values, defining equivalence classes. Unfortunately, the space spanning possible partitions is complex and grows rapidly as far the sample size increases.

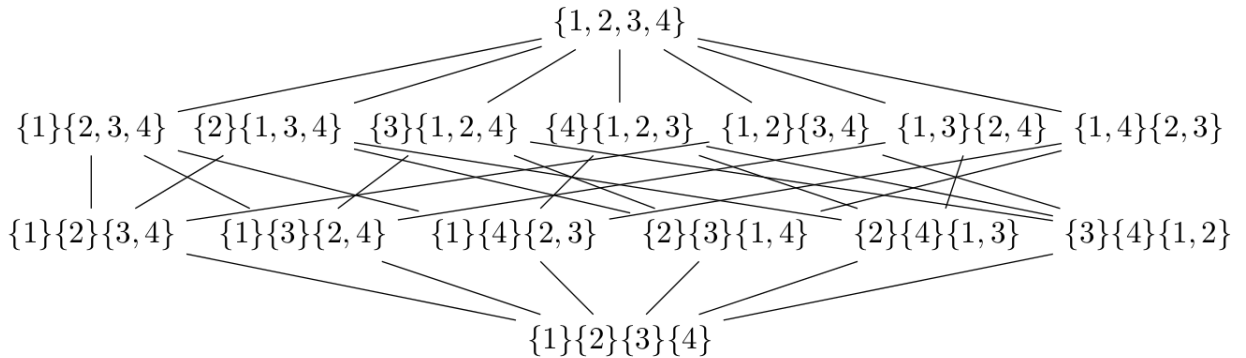


Figure 2: Lattice of all the possible partitions with $n = 4$ observations.

A partition is nothing but a way of dividing elements in subgroups. Here, we denoted a generic partition of n elements by ρ_n , which is then a set of blocks $\{A_1, \dots, A_k\}$, where the generic j th element $A_j = \{i \in \{1, \dots, n\} \text{ s.t. } s_i = j\}$, $j = 1, \dots, k$. The sets are disjoint, i.e. $A_i \cap A_j = \emptyset$,

for $i \neq j$. This means that an element can belong only to a single cluster. The union of all the blocks recover the set of observed indices, i.e. $A_1 \cup A_2 \cup \dots \cup A_k = \{1, \dots, n\}$. Hence, a partition describes exhaustively the allocation of all observations in the clusters. We further denote by n_j the size of each block, i.e. $n_j = |A_j| = \sum_{i=1}^n \mathbf{1}_{[s_i=j]}$, where the sum of all cluster sizes corresponds to the sample size, i.e. $n_1 + \dots + n_k = n$. Given n observations, we have

$$A_{n,k} = \frac{1}{k!} \sum_{j=0}^k (-1)^j \binom{k}{j} (k-j)^n$$

(Stirling number of the second kind) ways to partition n elements in k groups, and

$$B_n = \sum_{k=1}^n A_{n,k}$$

(Bell number) ways to partition n elements, which explodes as far n grows. Scan the entire partitions space is unfeasible in a reasonable time, as far as n grows. We need a different strategy to perform posterior inference on the clustering structure of our data.

n	1	2	3	4	5	6	7	8	9	10
B_n	1	2	5	15	52	203	877	4140	21147	115975

Table 1: Growth of the number of possible partitions as function of the sample size.

In practice, we produce a sample of the posterior distribution of interest resorting to the conditional distributions described in Proposition 1.4, obtaining a set of partitions which are representative of the latent cluster of the data. Ideally, we can exploit those full conditionals to derive a sampling strategies that perform posterior inference with mixture models, by iteratively sampling the quantities we need.

Input: $\alpha_1, \dots, \alpha_k$, parameter of $\pi(\theta_j^*)$. Initial values for $\theta_{1:k}^*$.

for $r = 1$ **to** R **do**

for $i = 1$ **to** n **do**

Sample the **cluster allocation** of the i th observation, with

$P(s_i = j \mid \mathbf{y}_{1:n}, \mathbf{w}, \theta_{1:k}^*) \propto w_j k(\mathbf{y}_i, \theta_j^*), \quad j = 1, \dots, k.$

end

for $j = 1$ **to** k **do**

Update the **cluster-specific parameters**, with

$\pi(\theta_j^* \mid \mathbf{y}_{1:n}, \mathbf{s}_{1:n}) \propto \pi(\theta_j^*) \prod_{i:s_i=j} k(\mathbf{y}_i, \theta_j^*), \quad j = 1, \dots, k.$

end

Update the **weights** with $\mathbf{w} \sim \text{Dir}(\alpha_1 + n_1, \dots, \alpha_k + n_k)$, where n_j is the size of the generic j th cluster.

end

The previous algorithm works quite intuitively, since at each iterations perform three samplings, associated with the three full conditionals of Proposition 1.4. Within each iteration,

a first for loop update the cluster allocation of each observation, conditionally on the rest. Hence, we are reallocating each observation to one of the possible k clusters, with probability proportional to each mixture component. A second for loop updates all the cluster-specific parameters, conditionally on the rest, hence also on the current allocation of the observations. Lastly, we update the weights associated to the mixture components.

We can be more computationally efficient with our clustering. An alternative approach, instead, looks at the marginal distribution of the cluster allocations, integrating out the distribution of the weights. Further, also the cluster-specific parameter can be marginalized for certain model choices, deriving an efficient sampling scheme that only updates cluster allocations. In this case, we have the following result.

Proposition 1.5. *Under the second augmentation, we have*

$$s_i \mid \mathbf{y}_i, \mathbf{w}, \boldsymbol{\theta}_{1:k}^* \stackrel{\text{ind}}{\sim} \text{Cat}(q_1, \dots, q_k), \quad i = 1, \dots, n,$$

where

$$q_j = P(s_i = j \mid \mathbf{y}_{1:n}, s_{-i}) \propto (\alpha_j + n_{j,-i}) \frac{m(\{y_i : s_i = j, i = 1, \dots, n\})}{m(\{y_\ell : s_\ell = j, \ell \neq i\})}$$

with $n_j = \sum_{i=1}^n \mathbf{1}_{[s_i=j]}$ is the number of observations assigned to the j th component of the mixture.

Proof. From the model specification, we have

$$\begin{aligned} L(\mathbf{y}_{1:n} \mid s_{1:n}, \boldsymbol{\theta}_{1:k}^*) &= \prod_{j=1}^k \prod_{i:s_i=j} k(\mathbf{y}_i, \boldsymbol{\theta}_j^*), & p(s_{1:n} \mid \mathbf{w}) &= \prod_{i=1}^n \prod_{j=1}^k w_j^{\mathbf{1}_{[s_i=j]}}, \\ \pi(\mathbf{w}) &= \frac{\Gamma(\alpha_1 + \dots + \alpha_k)}{\Gamma(\alpha_1) \dots \Gamma(\alpha_k)} \prod_{j=1}^k w_k^{\alpha_j-1}, & \pi(\boldsymbol{\theta}_{1:k}^*) &= \prod_{j=1}^k \pi(\boldsymbol{\theta}_j^*). \end{aligned}$$

Hence, the joint distribution of all the terms equals

$$\begin{aligned} \mathcal{L}(\mathbf{y}_{1:n}, s_{1:n}, \mathbf{w}, \boldsymbol{\theta}_{1:k}^*) &= \left(\prod_{j=1}^k \prod_{i:s_i=j} k(\mathbf{y}_i, \boldsymbol{\theta}_j^*) \right) \left(\prod_{i=1}^n \prod_{j=1}^k w_j^{\mathbf{1}_{[s_i=j]}} \right) \\ &\times \left(\frac{\Gamma(\alpha_1 + \dots + \alpha_k)}{\Gamma(\alpha_1) \dots \Gamma(\alpha_k)} \prod_{j=1}^k w_k^{\alpha_j-1} \right) \left(\prod_{j=1}^k \pi(\boldsymbol{\theta}_j^*) \right) \end{aligned}$$

We set $n_j = \sum_{i=1}^n \mathbf{1}_{[s_i=j]}$. We can integrate the distribution of the weights $\mathbf{w}$, obtaining

$$\begin{aligned} \mathcal{L}(\mathbf{y}_{1:n}, s_{1:n}, \boldsymbol{\theta}_{1:k}^*) &= \int_{\Delta_{k-1}} \mathcal{L}(\mathbf{y}_{1:n}, s_{1:n}, \mathbf{w}, \boldsymbol{\theta}_{1:k}^*) d\mathbf{w} \\ &= \left(\prod_{j=1}^k \prod_{i:s_i=j} k(\mathbf{y}_i, \boldsymbol{\theta}_j^*) \right) \left(\prod_{j=1}^k \pi(\boldsymbol{\theta}_j^*) \right) \frac{\Gamma(\alpha_1 + \dots + \alpha_k)}{\Gamma(\alpha_1) \dots \Gamma(\alpha_k)} \int_{\Delta_{k-1}} \prod_{j=1}^k w_k^{\alpha_j+n_j-1} d\mathbf{w} \\ &= \left(\prod_{j=1}^k \prod_{i:s_i=j} k(\mathbf{y}_i, \boldsymbol{\theta}_j^*) \right) \left(\prod_{j=1}^k \pi(\boldsymbol{\theta}_j^*) \right) \\ &\times \frac{\Gamma(\alpha_1 + \dots + \alpha_k)}{\Gamma(\alpha_1) \dots \Gamma(\alpha_k)} \frac{\Gamma(\alpha_1 + n_1) \dots \Gamma(\alpha_k + n_k)}{\Gamma(\alpha_1 + \dots + \alpha_k + n)}. \end{aligned}$$

We can integrate the previous also with respect to the cluster-specific parameter values, obtaining

$$\begin{aligned}
\mathcal{L}(\mathbf{y}_{1:n}, s_{1:n}) &= \int_{\Theta^k} \left(\prod_{j=1}^k \prod_{i:s_i=j} k(\mathbf{y}_i, \boldsymbol{\theta}_j^*) \pi(\boldsymbol{\theta}_j^*) \right) d\boldsymbol{\theta}_1^* \cdots d\boldsymbol{\theta}_k^* \\
&\times \frac{\Gamma(\alpha_1 + \cdots + \alpha_k)}{\Gamma(\alpha_1) \cdots \Gamma(\alpha_k)} \frac{\Gamma(\alpha_1 + n_1) \cdots \Gamma(\alpha_k + n_k)}{\Gamma(\alpha_1 + \cdots + \alpha_k + n)} \\
&= \prod_{j=1}^k \int_{\Theta} \prod_{i:s_i=j} k(\mathbf{y}_i, \boldsymbol{\theta}_j^*) \pi(\boldsymbol{\theta}_j^*) d\boldsymbol{\theta}_j^* \\
&\times \frac{\Gamma(\alpha_1 + \cdots + \alpha_k)}{\Gamma(\alpha_1) \cdots \Gamma(\alpha_k)} \frac{\Gamma(\alpha_1 + n_1) \cdots \Gamma(\alpha_k + n_k)}{\Gamma(\alpha_1 + \cdots + \alpha_k + n)} \\
&= \frac{\Gamma(\alpha_1 + \cdots + \alpha_k)}{\Gamma(\alpha_1 + \cdots + \alpha_k + n)} \prod_{j=1}^k \frac{\Gamma(\alpha_j + n_j)}{\Gamma(\alpha_j)} m(\{\mathbf{y}_i : s_i = j, i = 1, \dots, n\})
\end{aligned}$$

From the previous, the probability of allocating an observation to a specific cluster, conditionally on the cluster-specific parameters, can be identified by

$$\begin{aligned}
\mathcal{L}(\mathbf{y}_i, s_i \mid \mathbf{y}_{-i}, s_{-i}) &= \frac{\mathcal{L}(\mathbf{y}_{1:n}, s_{1:n})}{\mathcal{L}(\mathbf{y}_{-i}, s_{-i})} \\
&\propto \prod_{j=1}^k \frac{\Gamma(\alpha_j + n_{j,-i} + \mathbf{1}_{[s_i=j]})}{\Gamma(\alpha_j + n_{j,-i})} \frac{m(\{\mathbf{y}_i : s_i = j, i = 1, \dots, n\})}{m(\{\mathbf{y}_\ell : s_\ell = j, \ell \neq i\})} \\
&= \prod_{j=1}^k \frac{(\alpha_j + n_{j,-i})!}{(\alpha_j + n_{j,-i} - 1)!} \frac{m(\{\mathbf{y}_i : s_i = j, i = 1, \dots, n\})}{m(\{\mathbf{y}_\ell : s_\ell = j, \ell \neq i\})} \\
&= \prod_{j=1}^k (\alpha_j + n_{j,-i}) \frac{m(\{\mathbf{y}_i : s_i = j, i = 1, \dots, n\})}{m(\{\mathbf{y}_\ell : s_\ell = j, \ell \neq i\})}
\end{aligned}$$

and, consequentially,

$$P(s_i = j \mid \mathbf{y}_i, \boldsymbol{\theta}_{1:k}^*) \propto (\alpha_j + n_{j,-i}) \frac{m(\{\mathbf{y}_i : s_i = j, i = 1, \dots, n\})}{m(\{\mathbf{y}_\ell : s_\ell = j, \ell \neq i\})}.$$

□

The previous can be exploited to produce an efficient algorithm for model-based clustering, under the assumptions of having a prior distribution for the cluster-specific parameters $\pi(\boldsymbol{\theta})$ conjugate with the kernel function.

Input: $\alpha_1, \dots, \alpha_k$, parameter of $\pi(\boldsymbol{\theta}_j^*)$. Initial values for $\boldsymbol{\theta}_{1:k}^*$.

```

for  $r = 1$  to  $R$  do
  for  $i = 1$  to  $n$  do
    Sample the cluster allocation of the  $i$ th observation, with

      
$$P(s_i = j \mid \mathbf{y}_{1:n}, \mathbf{w}, \boldsymbol{\theta}_{1:k}^*) \propto (\alpha_j + n_{j,-i}) \frac{m(\{\mathbf{y}_i : s_i = j, i = 1, \dots, n\})}{m(\{\mathbf{y}_\ell : s_\ell = j, \ell \neq i\})}, \quad j = 1, \dots, k.$$


  end
end

```

Once we sampled a chain mimicking the posterior distribution of interest, we are usually interested at first about its convergence. Since the parameters of interest here are cluster allocations, we cannot produce directly a traceplot and the usual diagnostic quantities, but we resort to some functionals of such chain. Usually, we consider (i) the number of allocated components at each iteration, and (ii) the entropy of the partition at each iteration, to produce two sequence of values that we can plot and study.

Exercise 1.6. Let us consider the following data

```
set.seed(42)
y <- c(rnorm(50, -3, 1), rnorm(25, 3, 1))
```

We want to cluster those observations, assuming a mixture of univariate Gaussian, using a $N(\mu, \sigma^2)$ as kernel function. Further, we set as prior for the group-specific parameter

$$\begin{aligned}\mu \mid \sigma^2 &\sim N(\mu, \sigma^2/k_0), \\ \sigma^2 &\sim IG(a_0, b_0).\end{aligned}$$

- Show that the previous prior assumption is conjugate for the Gaussian likelihood, with $\mu, \sigma^2 \mid y_{1:n} \sim NIG(m_n, k_n, a_n, b_n)$, where

$$\begin{aligned}k_n &= k_0 + n, & m_n &= (k_0 m_0 + n \bar{y}) / (k_0 + n) \\ a_n &= a_0 + n/2, & b_n &= b_0 + \frac{1}{2} \left(\sum_{i=1}^n (y_i - \bar{y})^2 + \frac{k_0 n}{k_n} (\bar{y} - m_0)^2 \right).\end{aligned}$$

- implement a Gibbs sampler to sample from the posterior distribution of the latent partition in the data. Check the convergence of the algorithm by looking at the partition entropy at each iteration.

2 SUMMARIZING OUR POSTERIOR INFERENCE

We have a new problem, the so-called label switching. Once we produce a sample from the posterior distribution of interest, i.e. a sample of latent partitions $\rho_1, \dots, \rho_R$, we want to produce summaries of these sampled values, such as a single point estimate. Data are assumed to be exchangeable and the model is invariant with respect to permutation of the data or the components. Two observations can be both assigned to a specific component at a generic iteration r , and both to a different component at iteration $r + \ell$.

- From a clustering perspective, they belong to the same cluster, but from a label perspective they have different values.
- In our inferential procedure, we should take in account such situations.

Hence, looking at the most visited cluster or similar quantities, is not practical for this our inferential interest. We can approach the problem from a decision theory viewpoint. Let $L(\cdot, \cdot)$ denotes a loss function, having two partitions of n elements as arguments. The optimal

partition $\rho_n^* \in \mathbb{B}$ (or equivalently $(s_1^*, \dots, s_n^*) \in \mathbb{N}^n$) is the solution of

$$\begin{aligned}\rho_n^* &= \arg \min_{\hat{\rho}_n \in \mathbb{B}} \{E[L(\rho_n, \hat{\rho}_n) \mid \mathbf{y}_{1:n}]\} \\ &= \arg \min_{\rho_n \in \mathbb{B}} \left\{ \sum_{\hat{\rho}_n \in \mathbb{B}} L(\rho_n, \hat{\rho}_n) \Pr(\rho_n \mid \mathbf{y}_{1:n}) \right\},\end{aligned}$$

where ρ_n^* denotes the optimal partition, $\hat{\rho}_n$ the partition we are considering, ρ_n the partitions wrt we take the expectation, and $\Pr(\rho_n \mid \mathbf{y}_{1:n})$ is the posterior probability of ρ_n , given a set of observations $\mathbf{y}_{1:n}$. New problems here are that the space $\mathbb{B}$ is too large and we can choose $L(\cdot, \cdot)$ in several ways. Luckily, some choices lead to tractable expressions. A first loss function that we can consider is the 0 – 1 loss function, i.e.

$$L_{0-1}(\rho_n, \hat{\rho}_n) = \mathbf{1}_{[\rho_n \neq \hat{\rho}_n]}.$$

Such loss function is taking value 1 when all the blocks of ρ_n and $\hat{\rho}_n$ coincide, and 0 otherwise. Is not accommodating possible similarities in the partitions, all the partitions different from $\hat{\rho}_n$ are penalized in the same way.

Proposition 2.1. *The point estimate coincides with*

$$\arg \min_{\hat{\rho}_n \in \mathbb{B}} \left\{ \sum_{\rho_n \in \mathbb{B}} \mathbf{1}_{[\rho_n \neq \hat{\rho}_n]} \mathcal{L}(\rho_n = \hat{\rho}_n \mid \mathbf{Y}) \right\} = \arg \max_{\hat{\rho}_n \in \mathbb{B}} \{ \mathcal{L}(\rho_n = \hat{\rho}_n \mid \mathbf{Y}) \},$$

the maximum a posteriori of the distribution of ρ_n .

The proof of the previous results follow what we have seen in the first chapter regarding point estimates under 0 – 1 loss functions. Once we produce a sample of the posterior distribution, the most visited partition represent a point estimate under the previous loss function.

A more relaxed loss function is given by the Binder loss function.

$$L_B(\rho_n, \hat{\rho}_n) = \sum_{j < i} [C_1 \mathbf{1}_{[s_i = s_j]} \mathbf{1}_{[\hat{s}_i \neq \hat{s}_j]} + C_2 \mathbf{1}_{[s_i \neq s_j]} \mathbf{1}_{[\hat{s}_i = \hat{s}_j]}],$$

and by setting $C_1 = C_2$ it can be written as

$$L_B(\rho_n, \hat{\rho}_n) = \frac{1}{2} \left(\sum_{i=1}^{k_n} n_{i\bullet}^2 + \sum_{j=1}^{\hat{k}_n} n_{\bullet j}^2 - 2 \sum_{j=1}^{\hat{k}_n} \sum_{i=1}^{k_n} n_{ij}^2 \right).$$

It is based on the cluster frequencies, where

- k_n is the number of blocks in ρ_n
- $\hat{k}_n$ is the number of blocks in $\hat{\rho}_n$
- $n_{ij} = \sum_{\ell=1}^n \mathbf{1}_{[s_\ell = i]} \mathbf{1}_{[\hat{s}_\ell = j]}$ is the number of observations in the i -th block of ρ_n and in the j -th block of $\hat{\rho}_n$
- $n_{i\bullet} = \sum_{j=1}^{\hat{k}_n} n_{ij} = \sum_{\ell=1}^n \mathbf{1}_{[s_\ell = i]}$ is the number of observations in the i -th block of ρ_n

- $n_{\bullet j} = \sum_{i=1}^{k_n} n_{ij} = \sum_{\ell=1}^n \mathbf{1}_{[\hat{s}_\ell=j]}$ is the number of observations in the j -th block of $\hat{\rho}_n$

Proposition 2.2. *The solution of*

$$\rho_n^* = \arg \min_{\hat{\rho}_n \in \mathbb{B}} \left\{ \sum_{\rho_n \in \mathbb{B}} L_B(\rho_n, \hat{\rho}_n) \Pr(\rho_n \mid \mathbf{y}_{1:n}) \right\}$$

is given by the partition which, in binary representation, minimize the distance from the posterior similarity matrix $\mathcal{M}$, where

$$\mathcal{M}_{ij} = P(s_i = s_j \mid \mathbf{y}_{1:n}), \quad \text{and} \quad \mathbf{s}_{1:n}^* = \arg \min_{\hat{\mathbf{s}}_{1:n} \in \mathbb{N}_n} \left\{ \sum_{i < j} |\mathbf{1}_{[\hat{s}_i=\hat{s}_j]} - \mathcal{M}_{ij}| \right\}.$$

Proof. Starting from the posterior expectation of the loss function, we have

$$\begin{aligned} & \mathbb{E}[L_B(\rho_n, \hat{\rho}_n) \mid \mathbf{y}_{1:n}] \\ &= \sum_{\mathbf{s} \in \mathbb{N}_n} \sum_{i < j} [C_1 \mathbf{1}_{[s_i=s_j]} \mathbf{1}_{[\hat{s}_i \neq \hat{s}_j]} + C_2 \mathbf{1}_{[s_i \neq s_j]} \mathbf{1}_{[\hat{s}_i=\hat{s}_j]}] P(\mathbf{s}_{1:n} \mid \mathbf{y}_{1:n}) \\ &= \sum_{i < j} [C_1 \mathbf{1}_{[\hat{s}_i \neq \hat{s}_j]} P(s_i = s_j \mid \mathbf{y}_{1:n}) + C_2 \mathbf{1}_{[\hat{s}_i=\hat{s}_j]} P(s_i \neq s_j \mid \mathbf{y}_{1:n})] \\ &\stackrel{C_1=C_2}{\propto} \sum_{i < j} [(1 - \mathbf{1}_{[\hat{s}_i=\hat{s}_j]}) P(s_i = s_j \mid \mathbf{y}_{1:n}) + \mathbf{1}_{[\hat{s}_i=\hat{s}_j]} (1 - P(s_i = s_j \mid \mathbf{y}_{1:n}))] \\ &= \sum_{i < j} [P(s_i = s_j \mid \mathbf{y}_{1:n}) - 2\mathbf{1}_{[\hat{s}_i=\hat{s}_j]} P(s_i = s_j \mid \mathbf{y}_{1:n}) + \mathbf{1}_{[\hat{s}_i=\hat{s}_j]} \pm P(s_i = s_j \mid \mathbf{y}_{1:n})^2] \\ &\propto \sum_{i < j} [\mathbf{1}_{[\hat{s}_i=\hat{s}_j]} - 2\mathbf{1}_{[\hat{s}_i=\hat{s}_j]} P(s_i = s_j \mid \mathbf{y}_{1:n}) + P(s_i = s_j \mid \mathbf{y}_{1:n})^2] \\ &= \sum_{i < j} \left[\left(\mathbf{1}_{[\hat{s}_i=\hat{s}_j]} - P(s_j = s_j \mid \mathbf{y}_{1:n}) \right)^2 \right], \end{aligned}$$

which corresponds to the quantity of interest, where the last $\propto$ holds since we are interested only in quantities that relate to $\hat{\rho}_n$ or $\hat{\mathbf{s}}_{1:n}$, equivalently. $\square$

The Binder loss function, differently from the 0 – 1 loss function, is not penalizing in the same way the partitions different from the optimal one.

Example 2.3. A practical example: consider $n = 3$, and all the possible partitions are

$$\mathbb{B} = \{\{1, 2, 3\}, \{12, 3\}, \{1, 23\}, \{13, 2\}, \{123\}\}$$

Equivalently in a binary representation

$$\begin{aligned} W_1 &= \begin{bmatrix} 1 & & \\ 0 & 1 & \\ 0 & 0 & 1 \end{bmatrix}, & W_2 &= \begin{bmatrix} 1 & & \\ 1 & 1 & \\ 0 & 0 & 1 \end{bmatrix}, & W_3 &= \begin{bmatrix} 1 & & \\ 0 & 1 & \\ 0 & 1 & 1 \end{bmatrix}, \\ W_4 &= \begin{bmatrix} 1 & & \\ 0 & 1 & \\ 1 & 0 & 1 \end{bmatrix}, & W_5 &= \begin{bmatrix} 1 & & \\ 1 & 1 & \\ 1 & 1 & 1 \end{bmatrix} \end{aligned}$$

and $\mathcal{M} = \begin{bmatrix} 1 & & \\ 0.8 & 1 & \\ 0.2 & 0.2 & 1 \end{bmatrix}$, then we have

$$\begin{aligned} d(W_1, \mathcal{M}) &= 0.8^2 + 0.2^2 + 0.2^2 = 0.72; & d(W_2, \mathcal{M}) &= 0.08; & d(W_3, \mathcal{M}) &= 0.68; \\ d(W_4, \mathcal{M}) &= 0.68; & d(W_5, \mathcal{M}) &= 1.32, \end{aligned}$$

and the point estimate is given by R_2 .

Upon the previous, we can build an algorithm that produce a point estimate of the latent partition, under the Binder loss function, starting from a MCMC sample from the posterior distribution of interest.

Input: A sample of partitions from the posterior distribution $\{\rho^{(1)}, \dots, \rho^{(R)}\}$.
Produce an estimate of the posterior similarity matrix $\mathcal{M}$, where

$$[\mathcal{M}]_{i,j} = \frac{1}{R} \sum_{r=1}^R \mathbf{1}_{[s_i^{(r)} = s_j^{(r)}]}$$

for $r = 1$ **to** R **do**

 Reconstruct the binary matrix for the r th partition, with

$$[W_r]_{ij} = \begin{cases} 1 & : s_i^{(r)} = s_j^{(r)}, \\ 0 & : \text{otherwise} \end{cases}$$

 Compute the squared distance between $\mathcal{M}$ and W_r .

end

Return the partitions W_r that minimize the distance.

Exercise 2.4. Let us consider the previous data

```
set.seed(42)
y <- c(rnorm(50, -3, 1), rnorm(25, 3, 1))
```

with the same assumption of the previous example, i.e. using a $N(\mu, \sigma^2)$ as kernel function. Further, we set as prior for the group-specific parameter

$$\begin{aligned} \mu \mid \sigma^2 &\sim N(\mu, \sigma^2/k_0), \\ \sigma^2 &\sim IG(a_0, b_0), \end{aligned}$$

that a posteriori leads to $\mu, \sigma^2 \mid y_{1:n} \sim NIG(m_n, k_n, a_n, b_n)$, where

$$\begin{aligned} k_n &= k_0 + n, & m_n &= (k_0 m_0 + n \bar{y}) / (k_0 + n) \\ a_n &= a_0 + n/2, & b_n &= b_0 + \frac{1}{2} \left(\sum_{i=1}^n (y_i - \bar{y})^2 + \frac{k_0 n}{k_n} (\bar{y} - m_0)^2 \right). \end{aligned}$$

- Implement an R function to obtain a point estimate with the Binder loss function, starting from an MCMC output.
- Use the function to find the point estimate with the partitions sampled in the previous example.

3 THE MULTIVARIATE GAUSSIAN CASE

A common case while dealing with multivariate real-valued data is to consider a mixture of multivariate Gaussian distribution. Suppose we observe a sample $\mathbf{y}_1, \dots, \mathbf{y}_n$. Here, we want to identify clusters of observations where the generic $\mathbf{y}_i \in \mathbb{R}^p, i = 1, \dots, n$. In this framework, each term describing a cluster of data is given by a multivariate Gaussian component, with its own location and covariance matrix. The kernel function is then given by the density function of a multivariate Gaussian distribution, which is

$$f(\mathbf{y}_i | \boldsymbol{\mu}, \Sigma) = (2\pi)^{-p/2} |\Sigma|^{-1/2} \exp \left\{ -\frac{1}{2} (\mathbf{y}_i - \boldsymbol{\mu})^\top \Sigma^{-1} (\mathbf{y}_i - \boldsymbol{\mu}) \right\}.$$

Observations close to the centroid $\boldsymbol{\mu}$ belong to the cluster with high probability. Implicitly, we are looking at the Mahalanobis distance among observations belonging to the same cluster. The covariance matrix Σ drives the cluster dispersion and shape. To implement the algorithm we saw earlier in this chapter, we need first to set a distribution on the cluster-specific parameters.

- The location parameter $\boldsymbol{\mu}$ is a p -dimensional real-valued vector, hence we can set a multivariate Gaussian distribution.
- The covariance matrix is a $p \times p$ positive definite real-valued matrix. A possible distributional assumption for this parameter is given by the inverse-Wishart distribution.

Let Σ be a $p \times p$ positive definite real-valued matrix. We say that Σ is distributed as an inverse-Wishart distribution, with ν_0 degrees of freedom and scale parameter Λ_0 , if its density function corresponds to

$$\pi(\Sigma | \nu_0, \Lambda_0) = \frac{|\Lambda_0|^{\nu_0/2}}{2^{\nu_0 p/2} \Gamma_p(\nu_0/2)} |\Sigma|^{-(\nu_0+p+1)/2} e^{-\frac{1}{2} \text{tr}(\Lambda_0 \Sigma^{-1})},$$

where

$$\Gamma_p(\nu_0/2) = \pi^{p(p-1)/4} \prod_{j=1}^p \Gamma(\nu_0/2 + (1-j)/2)$$

is the multivariate gamma function. Note that, with the previous distributional assumption,

$$\mathbb{E}[\Sigma] = \frac{\Lambda_0}{\nu_0 - p - 1},$$

and

$$\text{Cor}(\Sigma_{[ij]}, \Sigma_{[kl]}) = \frac{2\Lambda_{0[ij]}\Lambda_{0[kl]} + (\nu_0 - p - 1)(\Lambda_{0[ik]}\Lambda_{0[jl]} + \Lambda_{0[il]}\Lambda_{0[jk]})}{(\nu_0 - p)(\nu_0 - p - 1)^2(\nu_0 - p - 3)}.$$

This might help us to specify the parameters ν_0 and Λ_0 .

Specifically, as prior assumption for the main parameter of the model, we consider an hierarchical specification of the form

$$\begin{aligned} \boldsymbol{\mu} | \Sigma &\sim N(\mathbf{m}_0, \Sigma/k_0), \\ \Sigma &\sim IW(\nu_0, \Lambda_0). \end{aligned}$$

which is called normal-inverse-Wishart distribution, where a priori we have

$$\begin{aligned}
\mathbb{E}_\pi[\boldsymbol{\mu}] &= \mathbb{E}_\Sigma \left[\mathbb{E}_\mu[\boldsymbol{\mu} \mid \Sigma] \right] = \mathbb{E}_\Sigma[\mathbf{m}_0] = \mathbf{m}_0, \\
\text{Var}_\pi(\boldsymbol{\mu}) &= \text{Var}_\Sigma \left(\mathbb{E}_\mu[\boldsymbol{\mu} \mid \Sigma] \right) + \mathbb{E}_\Sigma \left[\text{Var}_\mu(\boldsymbol{\mu} \mid \Sigma) \right] \\
&= \text{Var}_\Sigma(\mathbf{m}_0) + \mathbb{E}_\Sigma[\Lambda_0/k_0] = \frac{1}{k_0} \left(\frac{\Lambda_0}{\nu_0 - p - 1} \right),
\end{aligned}$$

and $\mathbb{E}_\pi[\Sigma]$, $\text{Var}_\pi(\Sigma)$ defined above.

Proposition 3.1. Assume the data are Gaussian distributed, with $\mathbf{y}_i \sim N(\boldsymbol{\mu}, \Sigma)$, for $i = 1, \dots, n$. Under the previous prior assumption $\boldsymbol{\mu} \mid \Sigma \sim N(\mathbf{m}_0, \Sigma/k_0)$ and $\Sigma \sim IW(\nu_0, \Lambda_0)$, we have

$$\begin{aligned}
\boldsymbol{\mu} \mid \Sigma &\sim N(\mathbf{m}_n, \Sigma/k_n), \\
\Sigma &\sim IW(\nu_n, \Lambda_n),
\end{aligned}$$

with

$$\begin{aligned}
k_n &= k_0 + n, & \mathbf{m}_n &= (k_0 \mathbf{m}_0 + n \bar{\mathbf{y}})/k_n \\
\nu_n &= \nu_0 + n, & \Lambda_n &= \Lambda_0 + \sum_{i=1}^n (\mathbf{y}_i - \bar{\mathbf{y}})(\mathbf{y}_i - \bar{\mathbf{y}})^\top + \frac{k_0 n}{k_n} (\bar{\mathbf{y}} - \mathbf{m}_0)(\bar{\mathbf{y}} - \mathbf{m}_0)^\top.
\end{aligned}$$

Hence, the normal-inverse-Wishart prior assumption is conjugate to the multivariate Gaussian likelihood.

Proof. The posterior distribution of $(\boldsymbol{\mu}, \Sigma)$ given a sample $\mathbf{y}_1, \dots, \mathbf{y}_n$ is equal to

$$\begin{aligned}
\pi(\boldsymbol{\mu}, \Sigma \mid \mathbf{y}_1, \dots, \mathbf{y}_n) &\propto L(\boldsymbol{\mu}, \Sigma; \mathbf{y}_{1:n}) \pi(\boldsymbol{\mu}, \Sigma) \\
&\propto \underbrace{|\Sigma|^{-1/2} |\Sigma|^{-(\nu_0+n+p+1)/2}}_A \exp \left\{ -\frac{1}{2} \sum_{i=1}^n (\mathbf{y}_i - \boldsymbol{\mu})^\top \Sigma^{-1} (\mathbf{y}_i - \boldsymbol{\mu}) \right. \\
&\quad \left. - \frac{1}{2} (\boldsymbol{\mu} - \mathbf{m}_0)^\top \left(\frac{\Sigma}{k_0} \right)^{-1} (\boldsymbol{\mu} - \mathbf{m}_0) - \frac{1}{2} \text{tr}(\Lambda_0 \Sigma^{-1}) \right\} \\
&= A \exp \left\{ -\frac{1}{2} \left[\sum_{i=1}^n \mathbf{y}_i^\top \Sigma^{-1} \mathbf{y}_i - 2 \boldsymbol{\mu}^\top \Sigma^{-1} \mathbf{s}_n + n \boldsymbol{\mu}^\top \Sigma^{-1} \boldsymbol{\mu} \right. \right. \\
&\quad \left. \left. + k_0 \boldsymbol{\mu}^\top \Sigma^{-1} \boldsymbol{\mu} - 2 k_0 \boldsymbol{\mu}^\top \Sigma^{-1} \mathbf{m}_0 + k_0 \mathbf{m}_0^\top \Sigma^{-1} \mathbf{m}_0 + \text{tr}(\Lambda_0 \Sigma^{-1}) \right] \right\}
\end{aligned}$$

We set $k_n = k_0 + n$ and $\mathbf{m}_n = (k_0 \mathbf{m}_0 + \mathbf{s}_n)/k_n$. Then we have

$$\begin{aligned}
&= A \exp \left\{ -\frac{1}{2} \left[-2 \boldsymbol{\mu}^\top \left(\frac{\Sigma}{k_n} \right)^{-1} \mathbf{m}_n + \boldsymbol{\mu}^\top \left(\frac{\Sigma}{k_n} \right)^{-1} \boldsymbol{\mu} \pm \mathbf{m}_n^\top \left(\frac{\Sigma}{k_n} \right)^{-1} \mathbf{m}_n \right. \right. \\
&\quad \left. \left. + \sum_{i=1}^n \mathbf{y}_i^\top \Sigma^{-1} \mathbf{y}_i + k_0 \mathbf{m}_0^\top \Sigma^{-1} \mathbf{m}_0 + \text{tr}(\Lambda_0 \Sigma^{-1}) \right] \right\} \\
&= A \exp \left\{ -\frac{1}{2} \left[(\boldsymbol{\mu} - \mathbf{m}_n)^\top \left(\frac{\Sigma}{k_n} \right)^{-1} (\boldsymbol{\mu} - \mathbf{m}_n) \right. \right. \\
&\quad \left. \left. + \text{tr} \left(\left(\sum_{i=1}^n \mathbf{y}_i \mathbf{y}_i^\top + k_0 \mathbf{m}_0 \mathbf{m}_0^\top - k_n \mathbf{m}_n \mathbf{m}_n^\top + \Lambda_0 \right) \Sigma^{-1} \right) \right] \right\}
\end{aligned}$$

$$\begin{aligned}
&= |\Sigma|^{-1/2} \exp \left\{ -\frac{1}{2} \left[(\boldsymbol{\mu} - \mathbf{m}_n)^\top \left(\frac{\Sigma}{k_n} \right)^{-1} (\boldsymbol{\mu} - \mathbf{m}_n) \right] \right\} \\
&\times |\Sigma|^{-(\nu_n + p + 1)/2} \exp \left\{ -\frac{1}{2} \left[\text{tr} \left(\left(\Lambda_0 + \sum_{i=1}^n (\mathbf{y}_i - \bar{\mathbf{y}})(\mathbf{y}_i - \bar{\mathbf{y}})^\top \right. \right. \right. \\
&\quad \left. \left. \left. + \frac{k_0 n}{k_n} (\bar{\mathbf{y}} - \mathbf{m}_0)(\bar{\mathbf{y}} - \mathbf{m}_0)^\top \right) \Sigma^{-1} \right) \right] \right\}
\end{aligned}$$

which is the kernel of a NIW distribution with updated parameters

$$\begin{aligned}
k_n &= k_0 + n, & \mathbf{m}_n &= (k_0 \mathbf{m}_0 + \mathbf{s}_n) / k_n, & \nu_n &= \nu_0 + n, \\
\Lambda_n &= \Lambda_0 + \sum_{i=1}^n (\mathbf{y}_i - \bar{\mathbf{y}})(\mathbf{y}_i - \bar{\mathbf{y}})^\top + \frac{k_0 n}{k_n} (\bar{\mathbf{y}} - \mathbf{m}_0)(\bar{\mathbf{y}} - \mathbf{m}_0)^\top
\end{aligned}$$

□

We now have all the quantities we need to implement a working algorithm to perform multi-variate clustering with Bayes, assuming a Gaussian kernel function.